

FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2024

TEST DATE: 17-12-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1.

B

Sol. $\frac{N_0}{2} = N_0 - kt^2$

$$kt^2 = \frac{N_0}{2}$$

$$t = \sqrt{\frac{N_0}{2k}} \text{ here } t \text{ is the time at which nuclei becomes half}$$

$$N = N_0 - kt^2$$

$$dN = -k(2t)dt$$

$$-\frac{dN}{dt} = 2k\sqrt{\frac{N_0}{2k}} = \sqrt{2kN_0}$$

2.

A

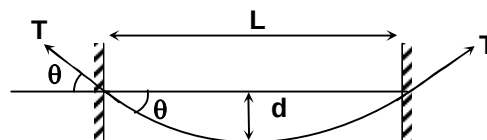
Sol. $2T \sin \theta = \mu Lg$

$$2T \left(\frac{d}{L/2} \right) = \mu Lg$$

$$T = \frac{\mu L^2 g}{4d}$$

$$\frac{T}{\mu} = \frac{L^2 g}{4d}$$

$$v_w = L \sqrt{\frac{g}{4d}}$$



3. C

Sol.
$$dU = \frac{1}{2} T \partial x \left(\frac{\partial y}{\partial x} \right)^2$$

$$\left(\frac{\partial y}{\partial x} \right) = \frac{2h}{\ell}$$

$$U = \frac{1}{2} (\mu c^2) \times \partial x \times \left(\frac{2h}{\ell} \right)^2 = \frac{2\mu c^2 h^2}{\ell^2} \int_0^\ell dx = \frac{2\mu c^2 h^2}{\ell}$$

4. A

Sol. Equilibrium

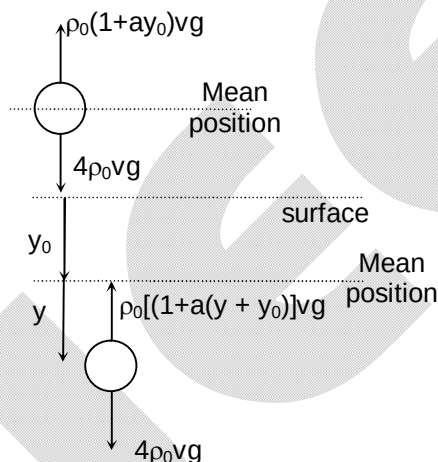
$$\rho_0(1+ay_0)vg = 4\rho_0vg$$

$$F_r = -\rho_0avgy$$

$$a_r = \frac{-\rho_0avg}{4\rho_0v} y$$

$$\Rightarrow a_r = -\frac{ag}{4} y$$

$$T = 2\pi \sqrt{\frac{4}{ag}} = \pi \text{ sec}$$



5. C

Sol.
$$\frac{\partial y}{\partial t} = 2A\omega \sin\left(\frac{\pi x}{L}\right) \cos \omega t$$

$$\frac{\partial y}{\partial t} = 2A\omega \sin\left(\frac{\pi x}{L}\right) = A\omega$$

$$\therefore \sin\left(\frac{\pi x}{L}\right) = \frac{1}{2}$$

$$\text{Separation, } \Delta x = L/3$$

6. C

Sol. Now,
$$\frac{1}{F_{\text{eff}}} = \frac{1}{f_m} - \frac{2}{f_0}$$

$$\frac{1}{10} = \frac{1}{f_m} + \frac{2}{30}$$

$$\frac{1}{f_m} = \frac{1}{10} - \frac{2}{30} = \frac{1}{30}$$

$$f_m = +30 \text{ cm}$$

$$R = +60 \text{ cm}$$

7. C

Sol. Total area = πa^2

$$\text{Force} = \frac{\pi a^2 l}{C}$$

8.

A

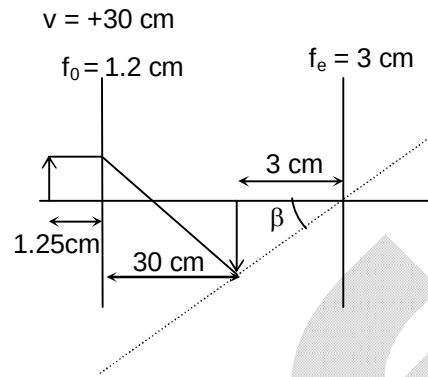
Sol. $f_0 = 1.2 \text{ cm}$, $f_e = 3 \text{ cm}$

$$\frac{1}{f_0} = \frac{1}{v_0} + \frac{1}{u_0}$$

$$\frac{1}{1.2} = \frac{1}{v_1} - \frac{1}{(-1.25)}$$

$$\Rightarrow \frac{1}{v} = \frac{5}{6} - \frac{4}{5} = \frac{1}{30}$$

$$\text{A.M.} = \left(\frac{v_0}{u_0} \right) \left(\frac{D}{u_e} \right) = \left(\frac{30}{1.25} \right) \left(\frac{25}{3} \right) = 200$$



9.

A

Sol. Let α is the angle between x-axis and plane of vibration

Law of malus

$$I = I_0 \cos^2 |\alpha - \theta|$$

10.

C

11.

C

Sol. $u = \frac{B^2}{2\mu_0}$

 $u \rightarrow$ energy per unit volume

$$\left[\frac{B^2}{\mu_0} \right] = [u] = \frac{[ML^2T^{-2}]}{[L^3]} = [ML^{-1}T^{-2}]$$

12.

B

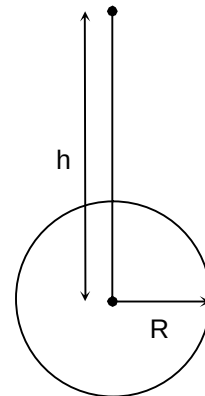
Sol.
$$-\frac{GMm}{R} + \frac{1}{2}mn^2V_e^2 = -\frac{GMm}{h}$$

$$-\frac{GMm}{R} + \frac{1}{2}n^2 \frac{2GMm}{R} = -\frac{GMm}{h}$$

$$\frac{n^2}{R} - \frac{1}{R} = \frac{-1}{h}$$

$$\frac{1}{h} = \frac{1-n^2}{R}$$

$$h = \frac{R}{1-n^2}$$



13.

C

Sol. With the increase in molecular weight, the rms speed of gas molecule decreases.

$$v_{rms} \propto \frac{1}{\sqrt{M}}$$

$$\frac{(v_{rms})_{H_2}}{(v_{rms})_{O_2}} = \sqrt{\frac{M_{O_2}}{M_{H_2}}} = \sqrt{\frac{16}{1}} = 4$$

$$(v_{rms})_{N_2} = 4v$$

14. A

 Sol. When any input is 'O' then current flows and $v_y = 0$ and when both input is 1
 No current flows and $v_y = 1$

15. D

 Sol. $Q = (B.E)_P - (B.E)_R$
 $= (104 + 116)(6.4) - (220)(5.6) = 176 \text{ MeV}$

16. D

 Sol. $\mu_r = 1.6, \epsilon_r = 6.4$
 $B = 4.5 \times 10^{-2}$
 $E = ?$

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}, V = \frac{1}{\sqrt{\mu \epsilon}}$$

$$\frac{C}{V} = \sqrt{\mu_r \epsilon_r} = \sqrt{1.6 \times 6.4}$$

$$\frac{E}{B} = V = \frac{3 \times 10^8}{\sqrt{1.6 \times 6.4}} = 9.3 \times 10^7 \text{ m/s}$$

$$E = 4.5 \times 10^{-2} \times 9.3 \times 10^7 = 4.2 \times 10^6$$

17. B

 Sol. $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$ and $v_{\text{mp}} = \sqrt{\frac{2RT}{M}}$, Thus $v_{\text{rms}} = \sqrt{\frac{3}{2}} v_{\text{mp}}$

18. C

 Sol. $\frac{I_1}{I_2} = \frac{1}{9}$

$$I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\frac{I_{\text{max}} + I_{\text{min}}}{I_{\text{max}} - I_{\text{min}}} = \frac{2(I_1 + I_2)}{4\sqrt{I_1}\sqrt{I_2}} = \frac{1}{2} \left[\frac{\sqrt{I_1}}{\sqrt{I_2}} + \frac{\sqrt{I_2}}{\sqrt{I_1}} \right] = \frac{5}{3}$$

19. A

 Sol. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\Rightarrow \frac{1}{3F_0} = \frac{1}{v} + \frac{1}{(-2F_0)}$$

$$\Rightarrow v = \frac{6F_0}{5}$$

$$\text{Image distance from object} = \frac{6F_0}{5} + 2F_0 = \frac{16F_0}{5}$$

Plane mirror

 Image distance from object = $2x$

$$2x = \frac{16F_0}{5} \Rightarrow x = \frac{8F_0}{5}$$

20. A

Sol. We use

$$hf_1 = \phi + \frac{1}{2}mv_1^2 \quad \dots(i)$$

$$\text{and } hf_2 = \phi + \frac{1}{2}mv_2^2 \quad \dots(ii)$$

using (i) and (ii) gives

$$v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$$

SECTION – B

21. 8

$$\text{Sol. } t = \frac{t_1 t_2}{t_1 + t_2} = \frac{6 \times 3}{9} = 2 \text{ hr}$$

6 hr is equal to three half lines

22. 3

Sol. Consider a small angular displacement Torque about the rotational axis

$$\text{Torque} = -\left(mg \frac{\ell}{2} \sin \theta\right)(\beta)$$

$$\text{Torque} = I\alpha$$

$$I = m \frac{(\ell \sin \theta)^2}{3} = m \frac{\ell^2 \sin^2 \theta}{3}$$

$$-\left(mg \frac{\ell}{2} \sin \theta\right)(\beta) = \frac{m(\ell \sin \theta)^2}{3} \alpha$$

$$\alpha = -\left(\frac{3g}{2\ell \sin \theta}\right)\beta$$

$$T = 2\pi \sqrt{\frac{2\ell \sin \theta}{3g}}$$

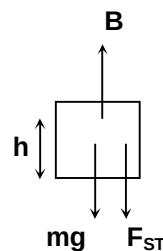
23. 1

$$\text{Sol. } mg + F_{ST} = B$$

$$mg + 4as = a^2 h \rho_w g$$

$$10 + 4 \times \frac{10}{4} = 10h$$

$$\Rightarrow h = 2$$



24. 4

$$\text{Sol. } a\sqrt{2gy} = (\pi x^2) \left(-\frac{dy}{dt}\right)$$

$$-\frac{dy}{dt} = \text{constant}$$

$$\Rightarrow y \propto x^4$$

25. 2

Sol. The central maxima

$$\frac{dy}{D} = \sqrt{d^2 + x^2} - x = x \left[1 + \frac{d^2}{2x^2} \right] - x = \frac{d^2}{2x} \quad (y \text{ is P.V. of central maxima from P})$$

$$y = \frac{Dd}{2x} \Rightarrow \frac{dy}{dt} = -\frac{Dd}{2x^2} \left(\frac{dx}{dt} \right)$$

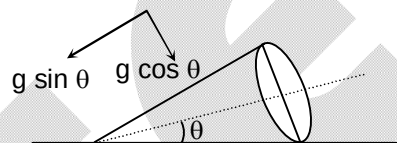
$$= \left(\frac{1 \times 0.01}{2 \times 0.5 \times 0.5} \right) \times (0.001) = 0.02 \text{ mm/s}$$

26. 5

$$\text{Sol. } F = [\rho(g \cos \theta)R] \pi R^2, \quad \cos \theta = \frac{R}{\sqrt{R^2 + H^2}} = \frac{1}{\sqrt{2}}$$

$$F = 1000 \times 10 \times \pi (1)^2$$

$$= 10^4 \times \frac{\sqrt{10}}{\sqrt{2}} = 10^4 \times \sqrt{5} \text{ N}$$



27. 2

$$\text{Sol. } \lambda = \frac{h}{mv}$$

28. 4

 Sol. Let ground state energy (in eV) be E_1 ,
Then from given condition,

$$E_{2n} - E_1 = 204 \text{ eV}$$

$$\frac{E_1}{4n^2} - E_1 = 204 \text{ eV}$$

$$E_1 \left(\frac{1}{4n^2} - 1 \right) = 204 \text{ eV}$$

$$\text{and } E_{2n} - E_n = 40.8 \text{ eV}$$

$$\frac{E_1}{4n^2} - \frac{E_1}{n^2} = 40.8 \text{ eV} \Rightarrow E_1 \times -\frac{3}{4n^2} = 40.8 \text{ eV}$$

From (i) and (ii)

$$\Rightarrow n = 2$$

Now from equation (ii)

$$E_1 = -\frac{4}{3}n^2(40.8)$$

$$\text{Or } E_1 = -217 \text{ eV}$$

$$\text{and } E_1 = -13.6 Z^2$$

Therefore

$$Z^2 = \frac{E_1}{-13.6} = \frac{-217.6}{-13.6} = 16$$

$$\Rightarrow Z = 4$$

29. 4

$$\text{Sol. We use } \frac{hc}{\lambda} = \phi + 3eV_0 \text{ and } \frac{hc}{2\lambda} = \phi + eV_0$$

Solving both equations

$$\frac{hC}{2\lambda} = 2\phi \Rightarrow \phi = \frac{hC}{4\lambda}$$

30. 4

Sol. $C = C_v + \frac{p}{n} \left(\frac{dV}{dT} \right), \frac{p}{n} \left(\frac{dV}{dT} \right) = \frac{R}{2}$

$$\Rightarrow \frac{T}{V^2} = \text{constant}$$

Chemistry

PART – B

SECTION – A

31. D

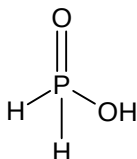
Sol. For strong electrolyte

$$\Lambda_m = \Lambda_m^{\circ} - A\sqrt{C}$$

 Λ_m° for weak electrolyte can be calculated by using λ_m° for individual ions.

32. B

Sol.


 reducing property of H_3PO_2 is due to P – H bond.

33. C

 Sol. $\text{Cu}(\text{BO}_3)_2 \Rightarrow$ Blue colour

34. D

 Sol. $\text{H}_2\text{A} + 2\text{KOH} \longrightarrow \text{K}_2\text{A} + 2\text{H}_2\text{O}$

$$\text{Moles of } \text{H}_2\text{A} = \frac{300 \times 0.2}{1000} = 60 \times 10^{-3}$$

$$\text{Moles of } \text{KOH} = \frac{700 \times 0.1}{1000} = 70 \times 10^{-3}$$

Limiting reagent is KOH

$$q = mS\Delta t$$

$$70 \times 10^{-3} \times 13.5 = 1000 \times 4.2 \times 10^{-3} \times \Delta t$$

$$\Delta t = 2.25 \times 10^{-1}$$

35. B

 Sol. K_H increases with increasing temperature. Higher the value of K_H , lower the solubility.

36. B

Sol.



mmol	8	2.4
		limiting

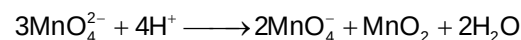
$$m \text{ moles of } \text{Ag}_2\text{CrO}_4 = 2.4$$

$$\text{moles of } \text{Ag}_2\text{CrO}_4 = 2.4 \times 10^{-3}$$

37. C

 Sol. $\text{Sc} > \text{Ti} > \text{Zn} > \text{Cu}$

38. A

 Sol. $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \longrightarrow 2\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{O}$

 MnO_4^{2-} green in colour and paramagnetic.

39. D
Sol. The magnetic properties of the actinoids are more complex than the lanthanoids.

40. D
Sol. $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ Pink
 $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ Purple
 $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ Green
 $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ Colourless

41. C
Sol. $\Delta T_f = K_f \frac{w}{MM \times W(\text{kg})}$
 $18.6 = 1.86 \times \frac{w}{62 \times 10}$
 $w = 6200 \text{ g}$

42. C
Sol. CuS is insoluble in yellow ammonium sulphide.

43. D
Sol. MnO_4^- is stronger oxidizing agent than $\text{Cr}_2\text{O}_7^{2-}$ in acidic medium.

44. A
Sol. Er is a lanthanoid.

45. A
Sol. $\text{Cu}^{2+} + \text{NaOH} \longrightarrow \text{Cu}(\text{OH})_2 \downarrow$
blue ppt.
 $\text{Pb}^{2+} + \text{NaOH} \longrightarrow \text{Pb}(\text{OH})_2 \downarrow \xrightarrow{\text{NaOH}} \text{Na}_2[\text{Pb}(\text{OH})_4]$
white ppt.
 $\text{Cd}^{2+} + \text{NaOH} \longrightarrow \text{Cd}(\text{OH})_2 \downarrow$
white ppt.
 $\text{Bi}^{3+} + \text{NaOH} \longrightarrow \text{Bi}(\text{OH})_3 \downarrow$
white ppt.

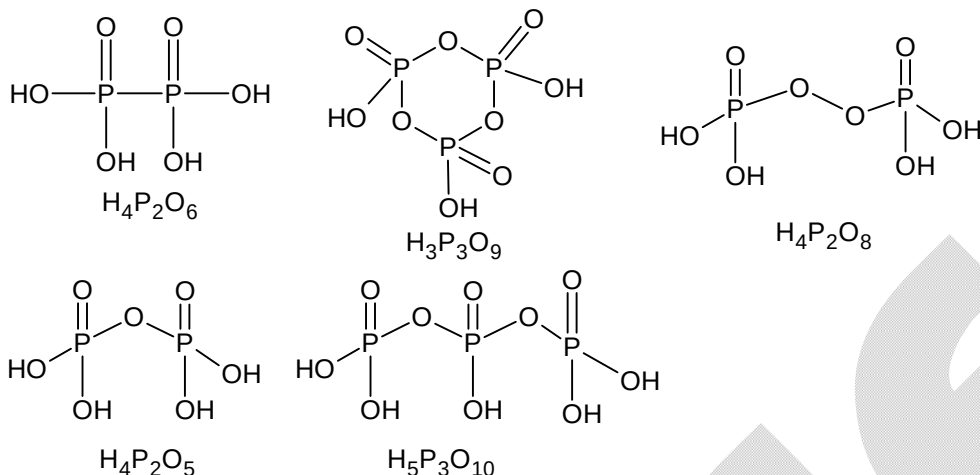
46. C
Sol. $[\text{Ni}(\text{CO})_4]$ and $[\text{FeCl}_4]^-$ have tetrahedral geometry
 $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ is a low-spin complex

47. D
Sol. $(\text{NH}_4)_2\text{CO}_3 \xrightarrow{\Delta} \text{NH}_3 + \text{H}_2\text{O} + \text{CO}_2$

48. A
Sol. $\text{HOCl} \longrightarrow \text{Cl}_2\text{O} + \text{H}_2\text{O}$

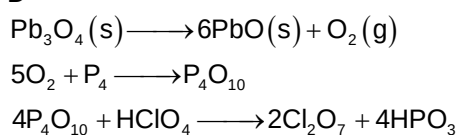
49. C

Sol.



50. D

Sol.



SECTION - B

51. 315

Sol.

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln \left[\frac{[\text{Ca}^{2+}]}{[\text{Cu}^{2+}]} \right]$$

$$3.185 = 3.20 - \frac{300}{2 \times 11500} \ln \left[\frac{[\text{Ca}^{2+}]}{[\text{Cu}^{2+}]} \right]$$

$$1.15 = \ln \frac{x}{1}$$

$$x = 3.15$$

52. 5

Sol.

$$(\Delta T_b) = iK_b m$$

$$= 2 \times 0.5 \times 0.2$$

$$= 0.4 \times 0.5$$

$$2\text{AgNO}_3 + \text{BaCl}_2 \longrightarrow 2\text{AgCl} \downarrow + \text{Ba}(\text{NO}_3)_2$$

0.1	0.1		
limiting			
0.1 - 0.1	0.1 - 0.05	0.1	0.05

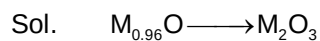
$$(\Delta T_b)_2 = (0.05 \times 3 + 0.05 \times 3) \times 0.5$$

$$= 0.15$$

$$(T_b)_2 - (T_b)_1 = 0.2 - 0.15$$

$$= 5 \times 10^{-2}$$

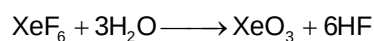
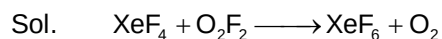
53. 88



$$n\text{-Factor} = \frac{(3 \times 0.96 - 2)}{0.96} \times 0.96$$

$$n\text{-Factor} = 0.88$$

54. 6



55. 25

Sol. $W_{Mg} = \frac{24 \times 100 \times 10^3}{10^6}$

$$= \frac{24}{10}$$

$$\text{Moles of Mg} = \frac{24}{10 \times 24} = \frac{1}{10}$$

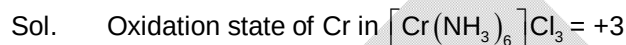
$$\text{Moles of } MgSO_4 \cdot 7H_2O = \frac{1}{10}$$

$$W_{MgSO_4 \cdot 7H_2O} = \frac{1}{10} \times 246$$

$$= 24.6$$

$$\approx 25$$

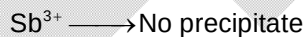
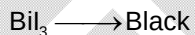
56. 3



$[Cr(NH_3)_6]Cl_3$ is an inner sphere complex. Total unpaired electrons is 3.



57. 1



58. 64

Sol. $\Delta S_{\text{system}} = nR \ln \frac{V_2}{V_1}$

$$= 20 \times 2 \ln \frac{1}{5}$$

$$= -64 \text{ cal K}^{-1}$$

$$\Delta S_{\text{surr}} = +64 \text{ cal K}^{-1}$$

59. 5

Sol. Cu^{2+} , Ag^+ , Zn^{2+} , Ni^{2+} and Cd^{2+} only.

60. 35

$$\text{Sol. } n = \frac{PV}{RT} = \frac{(714 - 14) \times 76}{760 \times 0.08 \times 10^3 \times 700} = \frac{1}{0.08} \times 10^{-4}$$

$$= \frac{1}{8} \times 10^{-2}$$

$$\% \text{ N} = \frac{1}{8} \times \frac{10^{-2} \times 28 \times 100}{0.1}$$

$$= 35$$

Mathematics**PART – C****SECTION – A**

61. B

Sol. $T_n = \frac{n^2 - n + 2}{2}$

62. B

Sol. $A^n = \begin{bmatrix} 1 & 2n & n\alpha + 8\sum_{r=1}^{n-1} r \\ 0 & 1 & 4n \\ 0 & 0 & 1 \end{bmatrix}$

63. D

Sol. Consider three points in 3-D
 $A(l, m, n)$, $B(x, y, z)$, $O(0, 0, 0)$

Now $\vec{OA} \cdot \vec{OB} = |\vec{OA}| |\vec{OB}| \cos(\angle AOB)$

$\Rightarrow \cos(\angle AOB) = \frac{lx + my + nz}{|\vec{OA}| |\vec{OB}|} = 1$

Also $\vec{OA} = \lambda \vec{OB}$

$\Rightarrow \frac{x}{l} = \frac{y}{m} = \frac{z}{n} = \frac{7}{6}$

64. A

Sol. $1, 2, 6, a, b$
 $(a - 4)^2 + (b - 4)^2 = 29$
 $a + b = 11$

65. D

Sol. $a\alpha^2 + b\alpha + c = 0$
 $\Rightarrow a\alpha^{n+1} + b\alpha^n + c\alpha^{n-1} = 0$
 $\Rightarrow a\beta^{n+1} + b\beta^n + c\beta^{n-1} = 0$
 adding : $aS_{n+1} + bS_n + cS_{n-1} = 0$
 now put $n = 2022$.

66. C

Sol. $\alpha + \beta = -(24)^{1/4}$, $\alpha \cdot \beta = 6^{1/2}$
 $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta \Rightarrow \alpha^2 + \beta^2 = 0$
 Now $\alpha^4 + \beta^4 + 2\alpha^2\beta^2 = 0 \Rightarrow \alpha^4 + \beta^4 = -12$
 $\alpha^8 + \beta^8 + 2\alpha^4\beta^4 = 144 \Rightarrow \alpha^8 + \beta^8 = 72$

67. A

Sol. Let $S = 1(\alpha_1 + \alpha_{2022}) + 2(\alpha_2 + \alpha_{2021}) + \dots + 2022(\alpha_{2022} + \alpha_1)$
 Also $S = 2022(\alpha_{2022} + \alpha_1) + 2021(\alpha_{2021} + \alpha_2) + \dots + 1(\alpha_1 + \alpha_{2022})$
 On adding
 $2S = 2023[2(\alpha_1 + \alpha_2 + \dots + \alpha_{2022})]$
 $S = -2023$

68. C

 Sol. Required no: $({}^5P_3) \times ({}^3P_2 + {}^3C_1) = 540$

69. D

 Sol. $P(x=r) = {}^nC_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{n-r}$

$$2P(x=5) = P(x=4) + P(x=6)$$

$$\Rightarrow 2 \cdot {}^nC_5 = {}^nC_4 + {}^nC_6$$

$$\Rightarrow n^2 - 21n + 98 = 0$$

$$\Rightarrow n = 7, 14.$$

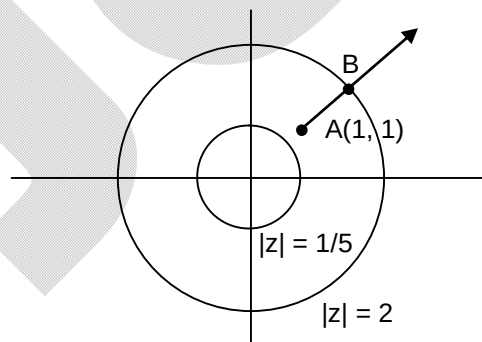
70. C

 Sol. $\alpha^4 + \beta^4 = ((\alpha + \beta)^2 - 2\alpha\beta)^2 - 2\alpha^2\beta^2$

$$= k^4 + \frac{1}{2k^4} + 2$$

$$\geq 2 + \sqrt{2}$$

71. B

 Sol. B is $(\sqrt{2}, \sqrt{2})$


72. B

 Sol. Clearly $\frac{1+z_2}{1+z_1}$ is purely real.

73. A

 Sol. $\Delta = 0 \Rightarrow t = 5$

$$I = \int_0^{-10} f(x) dx = \int_{-5}^{-10} f(x+5) dx + \int_{-5}^{-10} f(x) dx$$

$$= \int_{-5}^{-10} 2 dx = -10.$$

74. B

 Sol. $t_r = \sqrt{1 + \frac{1}{r^2} + \frac{1}{(r+1)^2}}$

$$= 1 + \frac{1}{r} - \frac{1}{r+1}$$

75. C

Sol. Required plane is passes through $\frac{\vec{a}_1 + \vec{a}_2}{2}$ and normal to the plane is parallel to $\vec{b}_1 \times \vec{b}_2$.

76. D

Sol. $(1+x+x^2+x^3)^n = \sum_{r=0}^{3n} a_r \cdot x^r$

Differentiate w.r.t. x and put x = 1.

77. D

Sol. Sphere touches the planes
 $x = 0, y = 0, z = 0$ and $3x + 4y + 12z = 24$
 let centre be (h, h, h)

$$\Rightarrow \left| \frac{3h+4h+12h-24}{\sqrt{3^2+4^2+12^2}} \right| = h$$

$$\Rightarrow h = 4, \frac{3}{4}$$

78. A

Sol. Required probability $\frac{\sum_{r=60}^{120} {}^{120}C_r}{2^{120}} = \frac{1}{2} + \frac{{}^{120}C_{60}}{(2)^{121}}$

79. D

Sol. Divide the numbers in 3 categories

- (i) ends with 0, 5
- (ii) ends with 2, 4, 6, 8
- (iii) ends with 1, 3, 7, 9

$$\therefore \text{Req. probability} = \left(\frac{4}{10}\right)^n$$

80. C

Sol. The equation of plane containing given lines is $x + 2y + 3z - 2 = 0$.

SECTION - B

81. 12

Sol. Image is $\frac{x-9}{18} = \frac{y+2}{-1} = \frac{z-1}{-10}$

82. 210

Sol. $\frac{C_1}{C_0} + 2 \cdot \frac{C_2}{C_1} + 3 \cdot \frac{C_3}{C_2} + \dots + n \cdot \frac{C_n}{C_{n-1}} = \frac{n(n+1)}{2}$

83. 4

Sol. $\frac{\frac{1}{2} |(\vec{a} - \vec{b}) \times (\vec{c} - \vec{b})|}{\frac{1}{2} |\vec{b} \times \vec{c}|} = 5$

Also $\vec{a} = -\alpha\vec{b} - \beta\vec{c}$

$\Rightarrow \alpha + \beta + 1 = 5.$

84. 468

Sol. Root of $x^3 - 9x^2 + kx - 24 = 0$ are in A.P.

$\Rightarrow 3\gamma = 9 \Rightarrow \gamma = 3, \alpha = 2, \delta = 4$

$\Rightarrow$ also, 2, 3, 4 are roots of $g(x)$

$\Rightarrow 2, \beta, 3, 4$ are in H.P.

$\Rightarrow \beta = 12/5$

$\Rightarrow f(5) \cdot g(5) = 468.$

85. 7

Sol. $|\text{adj adj adj } A| = |A|^{(n-1)^3}$ and $\text{adj adj adj } A = |A|^2 \text{ adj } A$

86. 105

Sol. $S_{12} - S_5 = a_6 + a_7 + a_8 + \dots + a_{12} = 7a_9$

87. 2024

Sol. $f(x) = 1 - x + x^2 - x^3 + \dots + x^{22} - x^{23} = a_0 + a_1(1+x) + \dots + a_{23}(1+x)^{23}$

$\Rightarrow f'(-1) = 2.1 + 3.2 + \dots + 23.22 = 2a_2.$

88. 1220

Sol. $f(x) - 5x = (x-1)(x-2)(x-3)(x-\lambda)$

$\Rightarrow f(7) + f(-3) = 1220.$

89. 2024

Sol. $z = a + ib \Rightarrow \bar{z} = a - ib$

$\Rightarrow z^{2022} = \bar{z}$

$\Rightarrow |z|^{2022} = |z|$

$\Rightarrow |z|(|z|^{2021} - 1) = 0$

$\Rightarrow z = 0 \text{ or } |z| = 1$

$\Rightarrow$ now $z^{2022} = \bar{z}$

$z^{2023} = z\bar{z} = 1$

$\Rightarrow$ 2023 value of z , total 2024 values.

90. 11851

Sol. $\left(1 + x + \frac{1}{x^2} + \frac{1}{x^3}\right)^{10} = \frac{(1 + x + x^3 + x^4)^{10}}{x^{30}}$

The term independent of x is coeff of x^{30} in $(1 + x + x^3 + x^4)^{10}$